



Harder Basic Arithmetic

Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 Solve for x : $\frac{1-2x}{1+x} \geq 1$

$$\begin{aligned} [x(1+x)^2] \quad (1-2x)(1+x) &\geq (1+x)^2, \quad x \neq -1 \\ (1+x)^2 - (1+x)(1-2x) &\leq 0 \\ (1+x)[(1+x) - (1-2x)] &\leq 0 \\ 3x(1+x) &\leq 0 \\ -1 < x \leq 0 \\ \text{OR } x &\in (-1, 0] \end{aligned}$$

2

2 Solve for x : $\frac{x}{x+1} > 2$

$$\begin{aligned} \text{Multiply both sides by } (x+1)^2 \\ x(x+1) &> 2(x+1)^2 \\ 2(x+1)^2 - x(x+1) &< 0 \\ (x+1)(2(x+1) - x) &< 0 \\ (x+1)(x+2) &< 0 \\ \therefore -2 < x < -1 \end{aligned}$$

2

3 Solve for x : $\frac{3}{x(2x-1)} \geq 1$

Multiply BOTH sides by the square of the denominator and solve the resulting inequality. 3

This resulting inequality is: $3x(2x-1) \geq x^2(2x-1)^2$, where $x \neq 0, \frac{1}{2}$.

Take all terms to one side and factorise if possible:

$$\begin{aligned} 0 &\geq x^2(2x-1)^2 - 3x(2x-1) \\ \therefore 0 &\geq x(2x-1)(x(2x-1) - 3) \\ \therefore 0 &\geq x(2x-1)(2x^2 - x - 3) \\ \therefore 0 &\geq x(2x-1)(2x-3)(x+1) \\ \left[-1, 0\right), \left[\frac{1}{2}, \frac{3}{2}\right] &\text{ OR } -1 \leq x < 0 \text{ and } \frac{1}{2} < x \leq \frac{3}{2} \end{aligned}$$

4 Solve: $\frac{3}{3x-1} \leq 5$

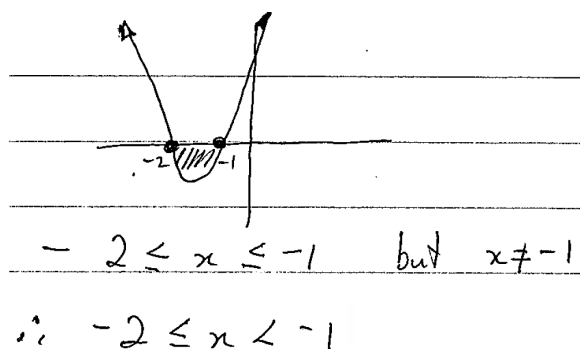
$$\begin{aligned} \frac{3}{(3x-1)} &\leq 5(3x-1)^2 \\ 3(3x-1) - 5(3x-1)^2 &\leq 0 \\ (3x-1)[3 - 5(3x-1)] &\leq 0 \\ (3x-1)[3 - 15x + 5] &\leq 0 \\ (3x-1)(8-15x) &\leq 0 \\ \frac{1}{3} &\leq x < \frac{8}{15} \end{aligned}$$

2

5 Solve: $\frac{1}{x+1} \leq -1$

2

$$\begin{aligned} \frac{1}{x+1} &\leq -1 \quad (x+1)^2 \\ x+1 &\leq -(x+1)^2 \\ (x+1)^2 + (x+1) &\leq 0 \\ (x+1)(x+1+1) &\leq 0 \\ (x+1)(x+2) &\leq 0 \end{aligned}$$



- 6 In trying to solve $\frac{x-2}{x-3} < \frac{4}{\sqrt{x-2}}$ over its natural domain in the set of real numbers, three students produce the following inequalities. 1

Student I $(x-2)\sqrt{x-2} < 4(x-3)$

Student II $\sqrt{(x-2)^3}(x-3) < 4(x-3)^2$

Student III $(x-2)^2(x-3) < 4(x-3)^2\sqrt{x-2}$

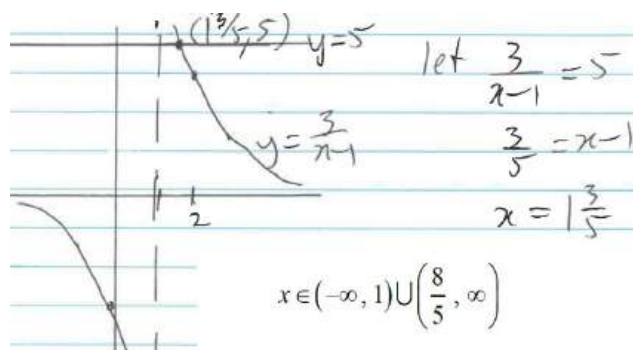
Which students are still on track to obtain the correct solution set?

- A. Just student II B. Student III only
C. Both students II and III but not student I D. All three students

Domain $x \neq 3$ and $x > 2$
 $\frac{x\sqrt{x-2}}{x(x-3)^2}$
 Student I is in error since sign of $x-3$ is indeterminate
 Student II is correct
 Student III is also correct, multiplication by extra factor of $\sqrt{x-2}$ on both sides is unnecessary but still gives a valid inequality hence (C)

- 7 The solution for the inequality $\frac{3}{x-1} < 5$ can be expressed as: 1

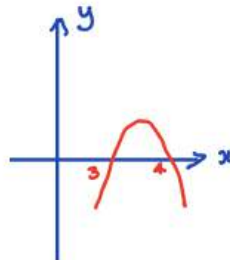
- A. $x \in \left[\frac{8}{5}, \infty\right)$ B. $x \in (-\infty, 1] \cup \left[\frac{8}{5}, \infty\right)$
 C. $x \in \left(\frac{8}{5}, \infty\right)$ D. $x \in (-\infty, 1) \cup \left(\frac{8}{5}, \infty\right)$



- 8 Solve: $\frac{4}{3x+1} \leq 1$ 3

$$\begin{aligned} \frac{4}{3x+1} &\leq 1 & x &\neq -\frac{1}{3} \\ 4(3x+1) &\leq (3x+1)^2 \\ 4(3x+1) - (3x+1)^2 &\leq 0 \\ (3x+1)[4 - (3x+1)] &\leq 0 \\ (3x+1)(-3x+3) &\leq 0 \\ (3x+1)(x-1) &\geq 0 \\ x &< -\frac{1}{3}, x \geq 1 \end{aligned}$$

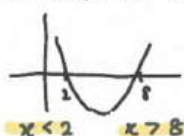
9 Solve: $\frac{x}{x-3} \leq 4$ for x .

$$\begin{aligned} \frac{x}{x-3} &\leq 4 & x \neq 3 \\ x(x-3) &\leq 4(x-3)^2 \\ x(x-3) - 4(x-3)^2 &\leq 0 \\ (x-3)[x-4(x-3)] &\leq 0 \\ (x-3)(x-4x+12) &\leq 0 \\ (x-3)(12-3x) &\leq 0 \\ 3(x-3)(4-x) &\leq 0 \\ \therefore (-\infty, 3) \cup [4, \infty) \\ x < 3 &\text{ or } x \geq 4 & (x \neq 3) \end{aligned}$$


10 For what values of x is $\frac{2x-3}{x-1} \leq 4$?

$$\begin{aligned} \frac{2x-3}{x-1} &\leq 4 & x \neq 1 \\ \frac{2x-3}{x-1} \times (x-1)^2 &\leq 4(x-1)^2 \\ (2x-3)(x-1) - 4(x-1)^2 &\leq 0 \\ (x-1)[2x-3-4(x-1)] &\leq 0 \\ (x-1)[2x-4x-3+4] &\leq 0 \\ (x-1)(-2x+1) &\leq 0 \\ (x-1)(2x-1) &\geq 0 \\ x &\leq \frac{1}{2}, \quad x > 1 & [x \neq 1] \end{aligned}$$

11 Solve for x : $\frac{3x}{x-2} < 4$

$$\begin{aligned} \frac{3x}{x-2} &< 4 \\ 3x(x-2) &< 4(x-2)^2 \\ 4(x-2)^2 - 3x(x-2) &> 0 \\ (x-2)[4(x-2) - 3x] &> 0 \\ (x-2)(4x-8-3x) &> 0 \\ (x-2)(x-8) &> 0 \\ x < 2 & \text{ or } x > 8 \end{aligned}$$


12 Solve for x : $\frac{4}{x+3} \geq x$

$$\begin{aligned} \frac{4}{x+3} &\geq x & x \neq -3 \\ 4(x+3) &\geq x(x+3)^2 \\ (x+3)[x(x+3)-4] &\leq 0 \\ (x+3)(x^2+3x-4) &\leq 0 \\ (x+3)(x+4)(x-1) &\leq 0 \\ \therefore x \leq -4, -3 < x \leq 1 &\text{ is the solution} \end{aligned}$$


13 Simplify $\frac{5^n + 5^{n+2}}{5^{n+1}}$

$$\frac{5^n + 5^{n+2}}{5^{n+1}} = \frac{5^n(1 + 5^2)}{5^n \times 5^1} = \frac{26}{5}$$

14 Solve: $|3x - 6| \geq 15$

$$\begin{aligned} 3x - 6 &\geq 15 & 3x - 6 &\leq -15 \\ 3x &\geq 21 & \text{ or } & 3x &\leq -9 \\ x &\geq 7 & & x &\leq -3 \end{aligned}$$



Harder Basic Arithmetic

Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 Solve: $\frac{1}{x+1} \leq -1$

$$\begin{aligned} (x+1)^2 \times \frac{1}{(x+1)} &\leq -1(x+1)^2 \\ (x+1) &\leq -(x^2+2x+1) \\ x+1 &\leq -x^2-2x-1 \\ x^2+3x+2 &\leq 0 \\ (x+2)(x+1) &\leq 0 \\ \therefore -2 \leq x < -1 \end{aligned}$$

3

2 Solve $(x^2 - 1)(x + 3) < 0$.

$$\begin{aligned} (x^2-1)(x+3) &< 0 \\ x^2-1=0 & \quad x+3=0 \\ x=\pm 1 & \quad x=-3 \\ \therefore x < -3 \text{ or } -1 < x < 1 \end{aligned}$$

2

3 The graphs of $y = \frac{1}{x-1}$ and $y = 2x + \frac{3}{2}$ are shown below. The functions

Intersect at y - values of -0.5 and 4 .

What is the solution to $2x + \frac{3}{2} < \frac{1}{x-1}$?

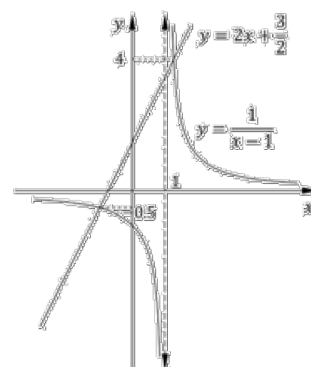
At the points of intersection:

$$2x + \frac{3}{2} = \frac{1}{x-1} \Rightarrow 2x = -2 \Rightarrow x = -1$$

$$2x + \frac{3}{2} = 4.5 \Rightarrow 2x = \frac{5}{2} \Rightarrow x = \frac{5}{4}$$

There is also the vertical asymptote at $x = 1$ to take into account. We need the x -values (not the y -values) for which the line is underneath the hyperbola.

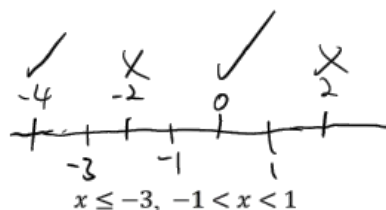
$$\left(-\infty, -1\right) \cup \left(1, \frac{5}{4}\right)$$



1

4 Solve the inequality $\frac{x+3}{x^2-1} \leq 0$.

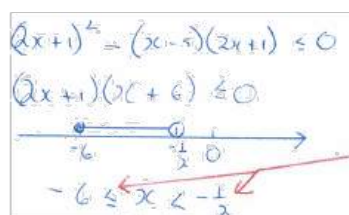
$$\begin{aligned} x^2 - 1 &\neq 0 \\ x^2 &\neq 1 \\ x &\neq \pm 1 \\ x + 3 &= 0 \\ x &= -3 \end{aligned}$$



3

5 Solve $\frac{x-5}{2x+1} \geq 1$.

$$\begin{aligned} \frac{x-5}{2x+1} &\geq 1 \quad x \neq -\frac{1}{2} \\ \frac{x-5}{2x+1} - 1 &\geq 0 \quad x \neq -\frac{1}{2} \\ \frac{(x-5)-(2x+1)}{2x+1} &\geq 0 \\ \frac{-x-6}{2x+1} &\geq 0 \end{aligned}$$



3

- 6 Solve the inequality $\frac{3}{x-1} < 2$.

$$\begin{aligned} \frac{3}{x-1} &< 2 \\ \frac{3}{(x-1)} &< 2 \frac{(x-1)^2}{(x-1)^2} \\ 3(x-1) &< 2(x-1)^2 \\ 3(x-1) - 2(x-1)^2 &< 0 \\ (x-1)(3-2(x-1)) &< 0 \\ (x-1)(3-2x+2) &< 0 \\ (x-1)(5-2x) &< 0 \end{aligned}$$

$x < 1$ or $x > 2.5$
(1 mark)

2

- 7 Express $\frac{1-x^{-1}}{x^{-1}-x^{-2}}$ in its simplest form.

$$\frac{1-x^{-1}}{x^{-1}-x^{-2}} = \frac{1-\frac{1}{x}}{\frac{1}{x}-\frac{1}{x^2}} = \frac{\frac{x-1}{x}}{\frac{x-1}{x^2}} = \frac{x-1}{x} \times \frac{x^2}{x-1} = x$$

2

- 8 Solve $x^4 - 3x^2 - 4 = 0$ for x .

Solve $x^4 - 3x^2 - 4 = 0$

Let $u = x^2$

$\therefore u^2 - 3u - 4 = 0$

[1]

$(u+1)(u-4) = 0$

i.e. $u = -1, 4$

$\therefore x^2 = -1$ (no real solution)

[1]

or $x^2 = 4$

$\therefore x = \pm 2$

[1]

3

- 9 Solve $4^x - 9(2^x) + 8 = 0$.

$$\begin{aligned} 4^x - 9(2^x) + 8 &= 0 \\ 2^{2x} - 9(2^x) + 8 &= 0 \\ \text{let } m &= 2^x \\ m^2 - 9m + 8 &= 0 \\ (m-8)(m-1) &= 0 \end{aligned}$$

$m = 8$	or	$m = 1$
$2^x = 8$		$2^x = 1$
$2^x = 2^3$		$2^x = 2^0$
$\therefore x = 3$		$\therefore x = 0$
$x = 0, 3$		

2

- 10 Given $2^x = 8^{y+1}$ and $9^y = 3^x$, find x and y .

$$\begin{aligned} \text{(ii)} \quad 2^x &= 8^{y+1} & 9^y &= 3^x \\ 2^x &= 2^{3(y+1)} & 3^{2y} &= 3^x \\ \Rightarrow x &= 3y+3, & & \\ x &= 2y & & \\ \Rightarrow 3y+3 &= 2y & & \\ y &= -3 & \therefore x &= -6 \end{aligned}$$

3

- 11 Solve $2|3x-1| - 1 \leq 7$

$$\begin{aligned} 2|3x-1| - 1 &\leq 7 \\ 2|3x-1| &\leq 8 \\ |3x-1| &\leq 4 \\ -4 &\leq 3x-1 \leq 4 \\ -3 &\leq 3x \leq 5 \\ -1 &\leq x \leq \frac{5}{3} \end{aligned}$$

3

- 12 Evaluate a, b and c if: $x^2 + 4x - 3 \equiv a(x+1)^2 + b(x+1) + c$

3

$$\begin{array}{l|l} a(x+1)^2 + b(x+1) + c & a + b + c = -3 \\ a(x^2 + 2x + 1) + bx + b + c & 1 + 2 + c = -3 \\ ax^2 + 2ax + a + bx + b + c & c = -6 \\ ax^2 + x(2a+b) + (a+b+c) & \checkmark \\ a = 1 & \text{Hence} \\ 2a + b = 4 & a = 1, b = 2, c = -6 \\ 2 + b = 4 & b = 2 \end{array}$$